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DGS-SLAM: A Visual Dense SLAM based on Gaussian Splatting in Dynamic Environments

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***Abstract*— Visual dense SLAM can facilitate pose estimation and map reconstruction for sensor carriers in unknown environments. However, in uncontrolled environments such as offices, shopping malls, and train stations, frequent occurrences of people walking back and forth or temporary movement of objects within the scene are common. Most existing visual dense SLAM systems do not account for these dynamic factors, leading to localization drift and map distortion. In this paper, we propose DGS-SLAM, a system capable of achieving robust localization and high-fidelity static map reconstruction in dynamic environments. We utilize semantic 3D Gaussians for scene representation, effectively eliminating interference from dynamic objects and refining the reconstruction of static background. We enhance the tracking accuracy and mapping quality of dense SLAM by using a distance distribution-based Gaussian pruning algorithm and implementing a coarse-to-fine tracking strategy with bundle adjustment and differentiable rendering. We perform qualitative and quantitative evaluations on two publicly available dynamic environment datasets. The results indicate that our method effectively reduces the interference caused by dynamic objects, enabling visual dense SLAM to maintain competitive tracking accuracy and mapping performance in dynamic environments.**

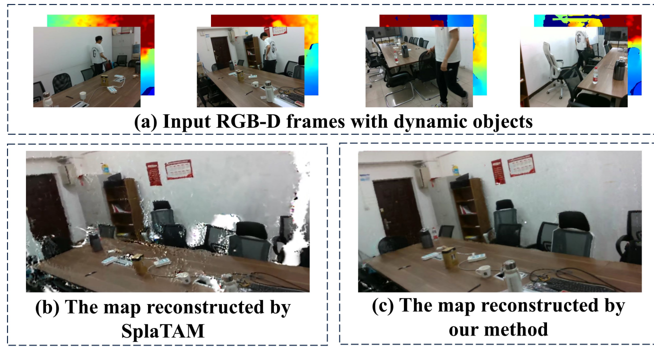

Fig. 1: Comparison between previous method with ours. (a) shows the input RGB-D frames; (b) illustrates the mapping results from SplaTAM [17], where many holes, scattered points, and artifacts are present; (c) presents the mapping results using our method, demonstrating high-fidelity mapping and the removal of interfering dynamic objects.

I. INTRODUCTION

In recent years, dense Simultaneous Localization and Mapping (dense SLAM) based on visual sensors has rapidly advanced [1]. It enables sensor carriers to achieve accurate self-localization while reconstructing realistic surrounding environments. This capability is crucial for applications such as augmented reality (AR), virtual reality (VR), and robotic navigation.

Most of visual dense SLAM approaches are based on the assumption of a static world, achieving good localization and mapping performance in the absence of dynamic objects. Traditional visual dense SLAM reconstructs scenes using direct methods, geometric modeling, or dense optical flow [2]–[5]. Neural Radiance Fields (NeRF)-based SLAM employs ray tracing with a Multi-Layer Perceptron (MLP) to render scenes and uses backpropagation to update MLP parameters and camera poses, achieving high-fidelity mapping [6]–[11].

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3D Gaussian Splatting (3DGS) [12] is a novel rendering technique that uses 3D Gaussians as differentiable explicit scene representation. It effectively models spatial geometric relationships and offers rendering quality comparable to NeRF, with significantly faster rendering speeds. Recent research has begun to integrate this technique with SLAM to achieve improved tracking and mapping performance [13]–[18].

However, the real world is not entirely static. In uncontrolled environments, such as office buildings, shopping malls, and stations, dynamic objects are often present. In these settings, visual dense SLAM frequently suffers from significant localization errors. This leads to severe distortions in the reconstructed maps, as shown in Fig. 1.

Some researchers have addressed this issue by incorporating semantic information, which mitigates the impact of dynamic interference on localization accuracy and map quality to a certain extent [19]–[27]. Nevertheless, they do not optimize pose and dense maps based on semantics after removing dynamic objects. There is still room for improvement in terms of rendering and reconstructing details of the scene in these approaches.

To tackle these challenges, we propose DGS-SLAM, a visual dense SLAM system that integrates 3D Gaussian Splatting with semantic attribute. Our system allows sensor carriers to achieve robust self-pose estimation and reconstruct reusable high-fidelity static maps even when navigating through unknown dynamic environments like crowded room or malls.

We propose a semantic 3D Gaussian scene representation. For highly dynamic object, we remove the corresponding Gaussians through a semantic-assisted Gaussian initialization. For static semantics, we enhance the pose and reconstruction accuracy of 3DGS-SLAM through differentiable rendering with semantic Gaussians, referred to as semantic-assisted optimization. We improve pose accuracy in the tracking process through a coarse-to-fine tracking strategy based on bundle adjustment and rendering loss. We also introduce a Gaussian pruning algorithm based on distance distribution in the mapping process, effectively removing artifacts and outliers.

Our method demonstrates improvements in tracking and mapping on the TUM-RGBD [28] and BONN dynamic datasets [29], validating the effectiveness of our SLAM. The main contributions of this paper can be summarized as follows.

- We propose DGS-SLAM, a robust SLAM system based on semantic-assisted initialization and optimization of 3D Gaussians, which provides robust localization and high-fidelity mapping in dynamic environments.
- We implement a semantic 3D Gaussian scene representation, allowing for the removal of dynamic objects and refined rendering of static background.
- We improve the tracking accuracy and mapping quality of dense SLAM by employing a coarse-to-fine tracking strategy and applying a distance distribution-based Gaussian pruning algorithm.

II. RELATED WORK

A. Visual Dense SLAM

Visual dense SLAM can help sensor carriers achieve self-localization and create a dense map of their surrounding environment while exploring unknown areas, which is of great significance for robotic applications. We summarize the advancements in this area as follows.

Traditional dense SLAM: classic dense SLAM methods achieve dense mapping through photometric error minimization [2], geometric modeling [3], and dense bundle adjustment [4], [5]. However, the resulting maps often lack realism and do not support novel view synthesis effectively. Therefore, researchers explore the integration of differentiable rendering with SLAM to achieve high-fidelity mapping and pose optimization.

NeRF-based SLAM: [6]–[11] employ implicit neural representation to track and map, improving the novel view synthesis capabilities. However, NeRF-based SLAM face a computational bottleneck from per-pixel ray marching, limiting their ability to balance high-fidelity mapping and efficiency. In contrast, 3D Gaussian splatting [12] introduces a differentiable rendering technique using 3D Gaussians as an explicit scene representation. It employs differentiable rasterization for fast rendering, achieving high-quality results by optimizing Gaussian attributes. This has made 3DGS a new focus of research.

3DGS-based SLAM: GS-SLAM [13] and Gaussian-SLAM [14] use RGB-D camera data as input and employ anisotropic 3D Gaussians for scene representation. These methods optimize camera poses and the attributes of 3D Gaussians in the map through backpropagation of color and depth losses. MonoGS [15] simplifies this approach by using isotropic 3D Gaussians for scene representation and imposing scale constraints to reduce reconstruction artifacts. It derives the explicit Jacobian determinant of camera poses in 3DGS-based SLAM, further enhancing the processing speed. Photo-SLAM [16] integrates traditional sparse visual SLAM with 3D Gaussian mapping capabilities, using visual odometry to quickly provide accurate poses and assist 3D Gaussians in efficient differentiable rendering and realistic map reconstruction. SplaTAM [17] employs contour-guided differentiable rendering, using high-quality 3D Gaussians for camera pose optimization and adaptively filling and refining the scene map based on reconstruction quality and Gaussian visibility. SGS-SLAM [18] incorporates semantic attributes in 3D Gaussians, providing additional constraints through semantic loss optimization during differentiable rendering, thus improving its ability to render details.

However, the aforementioned approaches exhibit a lack of robustness in dynamic environments. We addressed this deficiency by utilizing 3D Gaussian characteristic and semantic information.

B. Dynamic SLAM

In recent years, several studies have focused on addressing the challenges of visual SLAM in highly dynamic environments [19]–[25]. These approaches primarily aim to reduce localization errors by either removing or leveraging sparse features associated with dynamic objects. However, these methods mainly target localization in dynamic settings and do not contribute to a dense reconstruction of the scene.

Another line of work concentrates on mitigating the impact of dynamic objects during the dense mapping process in SLAM. Bârsan et al. [26] utilizes scene flow and semantic segmentation to distinguish the motion attributes of objects in the scene, separately reconstructing the static map and the instances within the scene. However, the reconstructed map lacks realism and the capability for novel view synthesis. [7], [30] have all noted the impact of dynamic objects on SLAM performance in their works. They enhance robustness in dynamic scenes by eliminating pixels with substantial depth rendering errors. Nevertheless, these approaches still lack robustness against significant dynamic disturbances, such as people walking. DN-SLAM [27] builds on Campos et al. [31] by removing potential dynamic objects through semantic segmentation and employs NeRF for dense mapping. RoDynSLAM [32] integrates optical flow masks and semantic masks within a NeRF-based SLAM framework to eliminate dynamic outliers.

Both DN-SLAM and RoDynSLAM primarily focus on the removal of dynamic objects, whereas our proposed method advances these approaches by further incorporating semantic-assisted differentiable rendering to optimize poses and dense

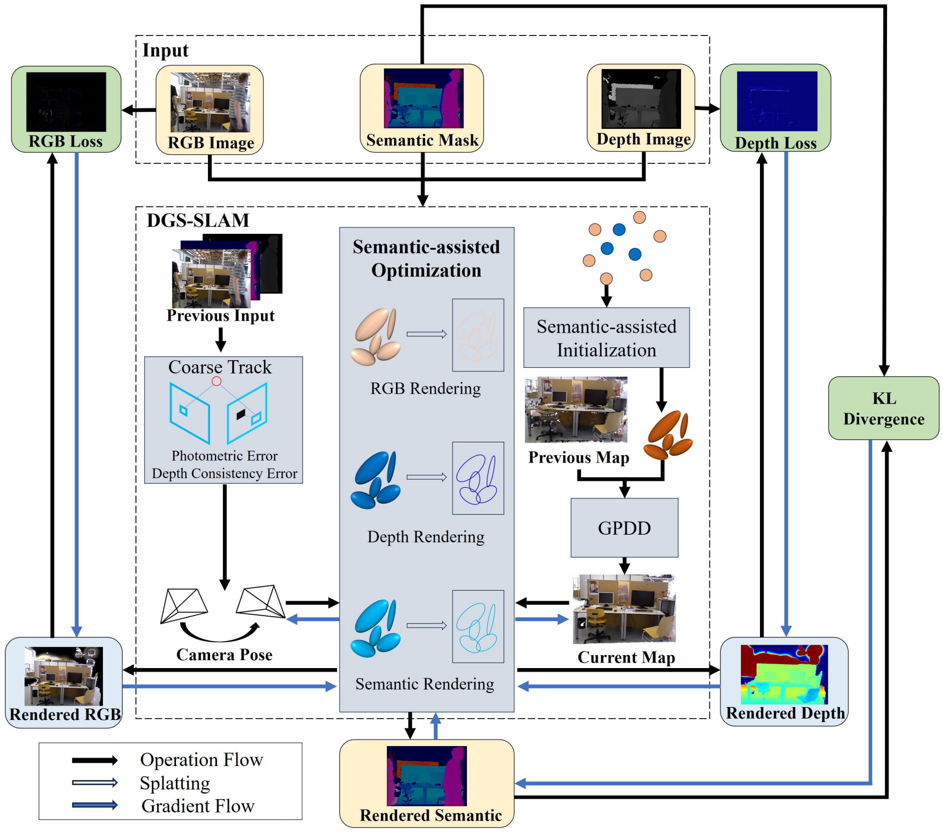

Fig. 2: System Overview. We employ semantic 3D Gaussians as an explicit and differentiable scene representation (III.A). The input semantic points with high dynamic semantics are excluded through semantic-assisted initialization (III.B). During the differentiable rendering process, we supervise using three signals: color, depth, and semantics. In the tracking phase, we obtain an initial pose estimation through photometric and depth consistency, followed by semantic-assisted optimization(III.C). For mapping, we apply Gaussian pruning based on distance distribution (GPDD) to reduce artifacts and outlier Gaussians in the scene. We also leverage semantic-assisted optimization to refine the Gaussians in the map (III.D).

map after dynamic object removal.

III. METHOD

A. Semantic 3D Gaussian Representation

To simplify computation while capturing multi-attribute features of the scene, we represent the scene as a set of isotropic 3D Gaussians:

$$\boldsymbol{G} = \{G_i : (\mu_i, r_i, o_i, c_i, s_i) | i = 1, \dots, N_g\}. \tag{1}$$

The parameters of each 3D Gaussian are defined as follows: N_g is the number of Gaussians; $\mu \in \mathbb{R}^{N_g \times 3}$ represents the mean of the Gaussians, indicating their positions in 3D space; r denotes the covariance of the Gaussians, which geometrically correspond to the radius of the isotropic Gaussians and control their volume; $o \in [0,1]^{N_g}$ is the opacity of the Gaussians; $c \in \mathbb{R}^{N_g \times 3}$ represents the color of the Gaussians, specified as RGB vectors; $s \in \mathbb{R}^{N_g \times m}$ denote the semantic category distribution vectors of 3D Gaussians, where m represents the number of semantic categories.

Each Gaussian can be represented by a Gaussian influence function for any point $x \in \mathbb{R}^3$ in the space as follows:

$$g^{3D}(x) = o \exp\left(-\frac{||x - \mu||^2}{2r^2}\right). \tag{2}$$

We project the 3D Gaussians onto the image plane, where their corresponding 2D positions and geometric attributes can be obtained using:

$$\mu^{2D} = K\frac{T_t \mu}{d}, \quad r^{2D} = \frac{fr}{d}, \quad d = (T_t \mu)_z, \tag{3}$$

where K denotes the camera's intrinsic matrix; T_t represents the camera pose matrix at time t; d indicates the depth information of 3D Gaussians; f is the camera's focal length; $(\cdot)_z$ extracts depth in the camera coordinate system, which corresponds to the depth value of the Gaussian.

During the differentiable rendering process, we perform rendering operations on the color, depth, and opacity attributes contained within the 3D Gaussians in the scene. In addition to these common rendering inputs, we also incorporate semantic distribution vectors into the optimization process, aiming to facilitate subsequent semantic-assisted initialization and optimization. The specific formula is shown as follows:

$$\mathcal{R}(v)=\sum_{i=1}^{n} v_i g^{2D}(x_i) \prod_{j=1}^{i-1}\left(1-g^{2D}(x_j)\right), \tag{4}$$

$$C=\mathcal{R}(c), \quad D=\mathcal{R}(d), \quad S=\mathcal{R}(s), \quad O=\mathcal{R}(o). \tag{5}$$

The Gaussian function $g^{2D}(x)$ is the corresponding representation of $g^{3D}(x)$ in 2D space. By applying α-blending, we can generate images using different attributes of the 3D Gaussians. Here, C, D, and S represent the images rendered based on the color c, depth d, and semantic attributes s of the 3D Gaussians, respectively; O denotes the silhouette image rendered from the opacity attribute o, which indicates the visibility of the Gaussians in the current scene.

B. Semantic-assisted Gaussian initialization

The task of the 3D Gaussian initialization process is to convert newly observed pixels into their corresponding 3D Gaussians. To remove dynamic elements from the scene, we determine whether to initialize the point cloud corresponding to each pixel as a 3D Gaussian based on the semantic attributes from the output of segmentation model.

Specifically, we use SegFormer [33] to obtain the semantic distribution of each pixel. Then we perform initialization for pixels with depth information. Following the dynamic characteristic score of common objects summarized in [20], we use prior knowledge to determine the motion attributes of the semantics. For dynamic semantic pixels, we do not initialize the corresponding point cloud. For static semantic pixels in the environment, we initialize them with an opacity O of 0.5, a mean μ as the coordinates of the corresponding point cloud in space, a covariance r as the ratio of the point cloud's measured depth d_m to the focal length f, a color c as the RGB value of the corresponding pixel and a semantic distribution vector s from segmentation model.

For static objects that undergo passive motion, such as a chair being moved by a person, we will detect and count the pixels that exceed ten times the median depth error during the semantic-assisted optimization process (which will be detailed in III.C and III.D). When a sufficient number of such pixels are identified within a particular semantic object, we will classify that object as dynamic and prevent its motion from affecting the tracking and mapping processes.

C. Coarse-to-fine Tracking

Initial Tracking: Typically, 3DGS SLAM methods utilize constant velocity models to estimate initial pose, followed by iterative differentiable rendering for pose optimization. We incorporate bundle adjustment with photometric and depth errors to perform preliminary pose optimization:

$$\min_{\xi} J(\xi)=\sum_{i=1}^{N_s}\left((1-\delta)\mathbf{e}_i^{pT}\mathbf{e}_i^{p}+\delta\mathbf{e}_i^{dT}\mathbf{e}_i^{d}\right), \tag{6}$$

$$^{p}e=\|I_{\text{src}}(x)-I_{\text{tgt}}(x')\|_1, \tag{7}$$

$$^{d}e=\|D_{\text{src}}(x)-D_{\text{tgt}}(x')\|_1, \tag{8}$$

$$x'=\pi\left(\exp(\xi^{\wedge})\pi^{-1}(x)\right). \tag{9}$$

Here, x represents a pixel in the source image, while x' denotes the corresponding pixel in the target image after being projected through the estimated pose. ξ denotes the pose to be optimized, N_s represents the number of pixel pairs from static region. ^{p}e denotes the photometric error of the corresponding pixel, while ^{d}e represents the depth error of the corresponding pixel. I_{src} and D_{src} refer to the grayscale and depth values of a pixel in the source image, respectively, and I_{tgt} and D_{tgt} refer to the grayscale and depth values in the target image. δ is a hyperparameter that determines the weighting of the errors and is set to 0.2 in our experiments. This bundle adjustment optimization approach allows for rapid acquisition of a good initial estimate for pose optimization based on differentiable rendering.

Semantic-assisted Pose Optimization: The camera pose will undergo further iterative optimization based on differentiable rendering. The loss function for differentiable rendering is defined as follows:

$$L_{track}=\sum_{p}\left(O(p)>t\right)\left(\lambda_d L_d(p)+\lambda_c L_c(p)+\lambda_s L_s(p)\right), \tag{10}$$

$$L_d(p)=\|D_r(p)-D_{gt}(p)\|_1, \tag{11}$$

$$L_c(p)=\|C_r(p)-C_{gt}(p)\|_1, \tag{12}$$

$$L_s(p)=S_{gt}(p)\log\frac{S_{gt}(p)}{S_r(p)}. \tag{13}$$

For each pixel p in the image, D_r, C_r, and S_r represent the rendered depth, color, and semantic images, while D_{gt}, C_{gt}, and S_{gt} correspond to the input depth image, the input color image, and the output probability distribution map of the semantic segmentation model. The parameter t is a visibility threshold, and λ_d, λ_c, and λ_s are weights that regulate the three types of errors. The depth error $L_d(p)$ and color error $L_c(p)$ are measured using L1 loss, while the semantic error $L_s(p)$ is quantified by the KL divergence between the distribution from model and the rendered distribution. During pose optimization, the parameters of the Gaussians in the map are kept fixed, and only those Gaussians with high visibility are used to ensure that excessive noise is not introduced.

D. Mapping

Semantic-assisted Map Optimization: We also optimize the Gaussians in the map through differentiable rendering. The loss function used in this process is similar to 10:

$$L_{map}=\sum_{p}\left(\lambda_d L_d(p)+\lambda_c L_c(p)+\lambda_s L_s(p)\right). \tag{14}$$

The meaning of each symbol is consistent with those in L_{track}, with the difference that during the map optimization process, camera poses are kept fixed, and all Gaussians are optimized.

(a) Input (b) ESLAM (c) SplaTAM (d) Ours

Fig. 3: Visualization of visual dense SLAM performance in dynamic environments.

TABLE I: Details of the data sequences used in the evaluation process.

Sequences	Description
walking-static	dynamic sequence with small camera motion
walking-halfsphere	dynamic sequence with large camera motion
crowd	sequence with a high presence of dynamic objects
balloon	sequence containing passively moving objects
desk	static sequence without dynamic objects

Gaussian Pruning based on Distance Distribution (GPDD): Due to dynamic interference or camera pose errors, the scene may contain isolated Gaussian outliers that can negatively impact mapping quality and camera tracking accuracy. To address this issue, we leverage the explicit nature of 3D Gaussians and propose a Gaussian pruning algorithm based on distance distribution. Specifically, we first calculate the average distance τ from each 3D Gaussian to its nearest K Gaussians. Next, we compute the mean μ_d and standard deviation σ_d of these τ values across all 3D Gaussians in the current map. Finally, any 3D Gaussians with τ values that fall within the intervals $\tau \in (-\infty, \mu_d - \alpha\sigma_d) \cup (\mu_d + \alpha\sigma_d, +\infty)$ are considered outliers and are subsequently removed.

IV. EXPERIMENTATION AND ANALYSIS

A. Experimental Setup

Datasets: We evaluate our method on two publicly available datasets, each containing several challenging RGB-D image sequences. As shown in I, we select five scene sequences from the TUM-RGBD dataset [28](walking-static, walking-halfsphere, desk) and the BONN dataset [29](crowd, balloon) for evaluation. Both the TUM-RGBD and BONN datasets provide RGB images, depth information, and ground truth camera poses. We employ SegFormer [33] to extract semantic information from images.

Implementation Details: All experiments are conducted on Intel Xeon E5-2678 v3, 32GB RAM and an RTX 4070 GPU. In our experiments, we set the parameter t to 0.99 as [17]. For the weights of the loss functions for 10 and 14, we assign $\lambda_d = 0.1$, $\lambda_c = 0.8$ and $\lambda_s = 0.1$. In the gaussian pruning process, we set K to 30 and α to 1. During both the tracking and mapping processes, the number of optimization iterations is set to 60.

B. Evaluation of Localization and Mapping

To validate the localization accuracy and reconstruction quality of our SLAM system in dynamic environments, we design a series of experiments focusing on both localization and mapping effectiveness.

Baselines and Details: We compare our method with the current state-of-the-art NeRF-SLAM(NICE-SLAM [7], ESLAM [9]), 3DGS-SLAM(SplaTAM [17], MonoGS [15]), and dynamic visual dense SLAM (DN-SLAM [27], RoDynSLAM [32]). To further demonstrate the contribution of each component in our system, we also conduct comparisons with several ablated variants of our method. The variants include: a system without the Gaussian pruning based on distance distribution (Ours wo GPDD), a system without the coarse-to-fine tracking strategy (Ours wo CFT), and a system without the incorporation of semantic attributes (Ours wo Sem). The results for DN-SLAM are derived from their published paper, while the other experimental results are based on their open-source code.

Metrics: For localization accuracy, we use the absolute trajectory error (ATE) as the metric, which is measured in centimeters. For reconstruction quality, we employ Depth L1 error, which is measured in centimeters. For rendering performance, we utilize PSNR, MS-SSIM, and LPIPS as evaluation metrics. In evaluating reconstruction and rendering performance, we adopt a semantic-aware evaluation approach similar to that in [26], calculating errors and similarities using pixels in static semantic regions and ignoring high-dynamic semantic regions.

Result Analysis: The quantitative experimental results related to reconstruction and rendering are presented in Table II, while the qualitative visual results are illustrated in 3. The results on localization performance are detailed in Table III. The results of the ablation study are summarized in Table IV. We highlight the best results for each metric in bold.

In terms of localization performance, significant improvements have been observed in DN-SLAM [27], RoDynSLAM [32], and our proposed method, as all these approaches incorporate specialized handling of dynamic objects. This contrasts to the SplaTAM [17], which does not take into account such considerations, resulting in comparatively inferior performance. In terms of mapping, existing NeRF-based SLAM and 3DGS-based SLAM methods exhibit substantial depth errors and suboptimal rendering quality. In contrast, our proposed approach ensures high-fidelity mapping performance in dynamic environments.

From the performance on the static sequence *desk*, we can see that our method maintains consistent performance with mainstream SLAM approaches in the static environment.

TABLE II: A comparative analysis of various SLAM systems in terms of mapping and rendering performance metrics. We highlight the best results in each metric in bold. Depth L1 error is measured in centimeters.

Sequences	Metric	NICE-SLAM	ESLAM	SplaTAM	MonoGS	Ours
walking-static	↓ Depth L1 (cm)	95.47	353.72	158.43	119.06	**7.12**
	↑ PSNR	18.80	15.27	26.29	15.83	**32.32**
	↑ SSIM	0.799	0.662	0.948	0.602	**0.985**
	↓ LPIPS	0.241	0.327	0.081	0.432	**0.029**
walking-halfshpere	↓ Depth L1 (cm)	116.93	314.79	46.46	126.46	**10.31**
	↑ PSNR	15.38	14.44	16.16	14.38	**25.25**
	↑ SSIM	0.659	0.640	0.723	0.520	**0.956**
	↓ LPIPS	0.339	0.361	0.309	0.503	**0.082**
crowd	↓ Depth L1 (cm)	110.07	285.28	27.37	83.30	**5.16**
	↑ PSNR	12.65	13.48	19.18	15.59	**27.96**
	↑ SSIM	0.541	0.657	0.795	0.646	**0.964**
	↓ LPIPS	0.484	0.300	0.221	0.530	**0.058**
balloon	↓ Depth L1 (cm)	29.71	237.03	19.13	75.99	**5.57**
	↑ PSNR	16.82	12.46	20.61	19.22	**26.07**
	↑ SSIM	0.739	0.609	0.794	0.768	**0.950**
	↓ LPIPS	0.287	0.352	0.217	0.359	**0.078**
desk	↓ Depth L1 (cm)	5.72	3.09	2.91	3.75	**2.83**
	↑ PSNR	16.36	17.23	23.14	**23.62**	23.19
	↑ SSIM	0.711	0.531	**0.907**	0.783	0.905
	↓ LPIPS	0.322	0.431	**0.154**	0.249	0.162

TABLE III: A comparative analysis of various SLAM systems in terms of the absolute trajectory error (ATE). ATE is measured in centimeters.

Sequences	SplaTAM	DN-SLAM	RoDynSLAM	Ours
walking-static	24.16	0.80	1.70	**0.34**
walking-halfsphere	95.01	2.60	5.60	**2.24**
crowd	52.58	2.50	13.17	**2.48**
balloon	38.23	3.00	7.90	**2.89**
desk	3.45	-	**2.54**	3.34

TABLE IV: Comparison of different ablation methods. Both ATE and Depth L1 error are measured in centimeters.

Method	↓ATE	↓Depth L1	↑PSNR	↑SSIM	↓LPIPS
Ours	**2.24**	**10.31**	**25.25**	**0.956**	**0.082**
Ours wo Sem	38.23	19.13	20.61	0.794	0.217
Ours wo CFT	2.56	12.82	24.50	0.948	0.096
Ours wo GPDD	2.53	12.57	24.46	0.948	0.093

From the ablation study results, it is evident that the semantic attributes of Gaussians have the most significant impact on the overall performance of the algorithm. This is primarily because they reduce the interference of dynamic outliers and introduce semantic constraints during the optimization process. The Gaussian pruning based on distance distribution (GPDD) method enhances mapping quality by pruning discrete Gaussians, which in turn contributes to improved localization accuracy. Additionally, the coarse-to-fine tracking (CFT) approach boosts localization precision by refining the initial optimization values during the tracking phase, while also positively influencing the reconstruction outcomes.

TABLE V: Comparison of runtime performance(s/frame).

Module	**ESLAM**	**SplaTAM**	**Ours**
Segmentation	-	-	0.03
Tracking	2.70	0.71	0.72
Mapping	2.73	0.82	0.86

C. *Time Evaluation*

We evaluate the runtime performance of the tracking and mapping components of our SLAM system on the TUM-RGBD dataset, with the results presented in Table V. Although our approach involves an additional semantic segmentation step compared to other methods, it achieves faster computation speed than NeRF-based SLAM. Compared to the current leading 3DGS-SLAM method, our approach enhances robustness in dynamic environments while still maintaining comparable runtime performance.

V. CONCLUSION

We present a 3DGS-based SLAM system that leverages semantic 3D Gaussians to reduce dynamic object interference and improve static background mapping. We also propose a coarse-to-fine tracking strategy incorporating bundle adjustment and differentiable rendering, along with a Gaussian pruning algorithm based on distance distribution to improve tracking and mapping performance. Experimental results on challenging datasets show our system achieves precise localization in dynamic scenes and reconstructs high-fidelity static maps.

We hope this work will further advance the environmental adaptability of visual dense SLAM. In the future, we plan to make lightweight improvements to our SLAM system to further enhance runtime performance.

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